MODULE *problem1*

EXTENDS *TLC*, *Integers*

VARIABLES *interested*, *pc*, *turn*

$processes \triangleq \{0, 1\}$

$TypeOK \triangleq$
$\quad \wedge \quad interested \in [processes \to \text{BOOLEAN}\ ]$
$\quad \wedge \quad turn \in processes$
$\quad \wedge \quad pc \in [processes \to \{\text{“T1”}, \text{“T2”}, \text{“T3A”}, \text{“T3B”}, \text{“E1”}\}]$

$Init \triangleq$
$\quad \wedge\ interested = [i \in processes \mapsto \text{FALSE}]$
$\quad \wedge\ turn \in processes$
$\quad \wedge\ pc = [p \in processes \mapsto \text{“T1”}]$

$T1(p) \triangleq$
$\quad \wedge\ pc[p] = \text{“T1”}$
$\quad \wedge\ pc' = [pc\ \text{EXCEPT}\ ![p] = \text{“T2”}]$
$\quad \wedge\ interested' = [interested\ \text{EXCEPT}\ ![p] = \text{TRUE}]$
$\quad \wedge\ \text{UNCHANGED}\ turn$

$T2(p) \triangleq$
$\quad \wedge\ pc[p] = \text{“T2”}$
$\quad \wedge\ pc' = [pc\ \text{EXCEPT}\ ![p] = \text{“T3A”}]$
$\quad \wedge\ turn' = 1 - p$
$\quad \wedge\ \text{UNCHANGED}\ interested$

$T3A(p) \triangleq$
$\quad \land\ pc[p] = \text{“T3A”}$
$\quad \land\ pc' = [pc \text{ EXCEPT } ![p] =$
$\quad\quad \text{IF } turn = p \text{ THEN “E1”}$
$\quad\quad \text{ELSE “T3B”}]$
$\quad \land\ \text{UNCHANGED } \langle turn, interested \rangle$

$T3B(p) \triangleq$
$\quad \land\ pc[p] = \text{“T3B”}$
$\quad \land\ pc' = [pc \text{ EXCEPT } ![p] =$
$\quad\quad \text{IF } interested[1 - p] \text{ THEN “T3A”}$
$\quad\quad \text{ELSE “E1”}]$
$\quad \land\ \text{UNCHANGED } \langle turn, interested \rangle$

$E1(p) \triangleq$
$\quad \land\ pc[p] = \text{“E1”}$
$\quad \land\ pc' = [pc \text{ EXCEPT } ![p] = \text{“T1”}]$
$\quad \land\ interested' = [interested \text{ EXCEPT } ![p] = \text{FALSE}]$
$\quad \land\ \text{UNCHANGED } turn$

$Mutex \triangleq$
$\quad \neg(pc[1] = \text{“E1”} \land pc[0] = \text{“E1”})$

$Next \triangleq$
$\quad \exists\, p \in processes :$
$\quad\quad \lor\ T1(p)$
$\quad\quad \lor\ T2(p)$
$\quad\quad \lor\ T3A(p)$
$\quad\quad \lor\ T3B(p)$
$\quad\quad \lor\ E1(p)$